A 72-operation relaxed-kernel consistency equation

Research supplement for ProveIt

23 October 2026

1 A 72-operation, degree-80 relaxed-kernel equation

The construction here has fifteen strictly positive integer unknowns, exact total degree 80, and 72 charged binary arithmetic operations. It saves one multiplication against the pinned degree-80 normalized P_{73} evaluator. This is a local tradeoff, not a global minimality claim: the inherited degree-155 equation still uses only 67 operations. The relaxed norm itself is prior art. Historical package 10 already has a degree-56, 74-operation equation using it; our companion P_{74} merely recovers that pair by a different grouping.

1.1 The explicit equation and its fixed coefficients

The witnesses, all in $\mathbb{Z}_{>0}$, are

$$J, F, \alpha, z, f, h, i, T, s, w, \tau, \eta, \zeta, y, \gamma.$$

The four fixed ports are the *complete* materialized Wang943 121-by-8801 tuple, with

$$B = 2^{1064921}, \quad \beta = B - 1, \quad K_0 = D_C + BD_R, \quad M_C, \quad M_F = M_{F,\text{native}} + B - 1.$$

The files `COEFFICIENT_BINDING.json` and `coefficients/` specify their exact bytes and hashes. The complete native adapter proof is retained as `upstream_verified/COEFFICIENT_PROOF.md`. No component is taken from the different complement-packed coefficient tuple.

Define polynomial abbreviations, not additional unknowns:

$$\begin{aligned}
q &= \beta J + 1, & C &= q - F - \alpha, & X &= wq,\\
Y &= sq^3, & E &= XY, & k &= \eta + \zeta,\\
c &= kY + \eta, & a &= E + Y, & A &= a + 2,\\
H &= 4a + 3, & \Delta &= a^2 + H = A^2 - 1, & D &= X + ac + \gamma H.
\end{aligned}$$

The remaining abbreviations are

$$\begin{aligned}
G &= (q-1)(q-F) + \alpha, & R &= G(q^2-1) + (M_C + qM_F)J,\\
V &= c(Tf - 1) - Rf^2, & I &= ic^2, \qquad K_{\text{aux}} = \Delta(f^2 - 1).
\end{aligned}$$

Set

$$\begin{aligned}
N_0 &= \tau^2 - (XY^2)(XY^2+1)k^2, & N_1 &= D^2 - \Delta c^2,\\
N_a &= K_{\text{aux}}(V^2 - y^2) + y^2, & N_k &= k - hE - R,\\
N_t &= (K_0 + w)C + (q - F) - z(q-1), & S &= I^2 - K_{\text{aux}}.
\end{aligned}$$

Our equation is

$$\boxed{(N_0N_1-1)^2+(N_aN_kN_t-1)^2+S^2=0.} \tag{1}$$

The companion is

$$(N_0-1)^2+(N_a-1)^2+S^2+(N_1N_kN_t-1)^2=0. \tag{2}$$

The literal files `p72_relaxed.json` and `p74_relaxed.json` contain the full division-free evaluators. Only fixed nonnegative exponents occur.

1.2 What must be proved after relaxing the norm

The new condition is

$$(ic^2)^2=\Delta(f^2-1),$$

not the normalized condition $f^2-\Delta(ic^2)^2=1$. These individual equations are not equivalent at the same coordinates. In completeness, the old multiplier is scaled by Δ; in soundness, a rank argument must first recover integral Pell coordinates. Consequently the existing Lean same-coordinate theorem for normalized P_{75} cannot be cited as a certificate for this polynomial.

At a zero of (1), $K_{\mathrm{aux}}=I^2$ and each of N_0,N_1,N_a,N_k,N_t is an integer unit. The negative-Pell descent for $v(v+1)$, with $v=XY^2>1$, excludes $N_0=-1$. Since $\Delta=(a+2)^2-1\equiv 0$ or $3 \pmod 4$, $N_1\neq -1$. Since K_{aux} is a square, $N_a\equiv y^2$ or $V^2 \pmod 4$, so $N_a\neq -1$. Thus

$$N_0=N_1=N_a=1, \qquad N_k=\epsilon, \quad N_t=\nu, \qquad \epsilon\nu=1, \quad \epsilon,\nu\in\{-1,1\}. \tag{3}$$

The same conclusion follows from (2). Neither remaining sign is assumed positive at this point.

Weak transport implies $C\geq 0$, not yet $C>0$. The authentic mask ranges then give

$$q(q^2-1)<R<q^4-q^3+q^2-1<q^4-2, \qquad R-2>3q+1, \qquad E>R+2, \qquad a>R+2.$$

Let $P_0=2XY^2+1$. Ordinary Pell classification yields

$$\tau=\chi_{P_0}(n), \quad k=2\psi_{P_0}(n), \qquad D=\chi_A(p), \quad c=\psi_A(p).$$

Since $P_0\equiv 1 \pmod E$, $2n=R+\epsilon+bE$ for some $b\geq 0$. Thus $n\geq 24$. Moreover $P_0>A$ and $c>k$ give $p>n$; Pell growth gives

$$c>A\Delta^2, \qquad c>2p, \qquad c>2R. \tag{4}$$

No auxiliary index or decoded tableau is used for these bounds.

1.3 The inherited relaxed-rank lemma, proved explicitly

Suppose $c=\psi_A(p)>A\Delta^2$ and $(ic^2)^2=\Delta(f^2-1)$. Then there is $m>0$ with

$$f=\chi_A(m), \qquad p\mid m, \qquad c\mid m, \qquad ic^2=\Delta\psi_A(m). \tag{5}$$

Here is the full rank argument. Write $\Delta=d_0s_0^2$ with $d_0>1$ squarefree and choose the least positive integer-coefficient unit $F+y_0\sqrt{d_0}$. By division by powers of this least unit, every positive integer-coefficient norm-one unit is its power. For some $e\geq 1$,

$$A=\chi_F(e), \quad S_0=\psi_F(e), \quad s_0=y_0S_0, \quad \Delta=(F^2-1)S_0^2, \quad \psi_F(ep)=S_0c.$$

The relaxed equation implies $s_0\mid ic^2$ and $d_0\mid ic^2/s_0$, so it too determines an integral norm-one unit. For some $t>0$,

$$f=\chi_F(t), \qquad ic^2=(F^2-1)S_0\psi_F(t).$$

Put $L = (F^2-1)S_0 \le \Delta$ and $g = \gcd(t, ep)$. Strong divisibility of the Pell sequence gives

$$c\gcd(c,\Delta) = \gcd(c^2, L\psi_F(ep)) \mid L\psi_F(g), \qquad \psi_F(g) \ge c/\Delta.$$

If $g < ep$, then $2g \le ep$ and doubling gives

$$2\psi_F(g)^2 < \psi_F(2g) \le \psi_F(ep) = S_0 c < Ac,$$

contrary to $c^2/\Delta^2 > Ac$. Hence $ep \mid t$. Write $t = em$; composition proves $p \mid m$ and the first and last claims in (5).

For the remaining divisibility, write $m = pb$. The binomial expansion gives

$$\psi_A(pb)/c \equiv bD^{b-1} \pmod{c}.$$

Since $c^2 \mid \Delta\psi_A(m)$ and $\gcd(D, c) = 1$, we have $c \mid \Delta b$. Also $c \equiv pA^{p-1} \pmod{\Delta}$, so $g_0 = \gcd(c, \Delta) = \gcd(p, \Delta)$ divides p. Thus $c/g_0 \mid b$ and $c \mid pb = m$, as required. This proof does not assert the stronger, generally unavailable condition $c^2 \mid \psi_A(m)$.

1.4 Quotient elimination is still sound

The rank lemma must be used *before* restoring the two deleted quotients. Set $\rho = ic^2$. Since $p \mid m$, we have

$$c^2 \mid f^2 - 1, \quad f^2 \ge 1 + \Delta c^2, \quad f > 2c, \quad \rho^2 = \Delta(f^2-1), \quad c \mid \rho, \quad m \ge c > 2p.$$

As $R < a < \Delta$,

$$\rho^2 - (Rf^2 + c) = (\Delta - R)f^2 - \Delta - c \ge f^2 - \Delta - c > 0.$$

The auxiliary equation is $\rho^2 V^2 - (\rho^2 - 1)y^2 = 1$. If $|V| > 1$, it implies $|V| \ge 2\rho^2 - 1$. But $T > 0$ gives $V > -(Rf^2 + c) > -\rho^2$, excluding negative values with $|V| > 1$. The cases $V = \pm 1$ contradict $V \equiv -c \pmod{f}$ and $f > 2c > 4$. Thus $V > 0$.

Now the proof-only expressions

$$o = cT - Rf, \qquad j = Tf - R(f^2-1)/c - 1$$

are integers, and $of = V + c > 0$, $jc = V + R > 0$. Consequently $o, j > 0$ and

$$V = of - c = jc - R.$$

There is no division in the polynomial evaluator. In particular, the old normalized formula for j in terms of $\Delta i^2 c^3$ must not be used with the new multiplier.

The retained signed auxiliary step-down applies with exactly these hypotheses and (4). To expose its interface, write $\rho V = \chi_\rho(\ell)$ with $\ell = 2u + 1$ odd and $\chi_Z(2u+1) = ZQ_u(Z^2)$. The identities

$$Q_u(1 - A^2) = (-1)^u \psi_A(\ell), \qquad Q_u(0) = (-1)^u \ell$$

give $(-1)^u\psi_A(\ell) \equiv -c \pmod{f}$ and $(-1)^u\ell \equiv -R \pmod{c}$. Squaring the first and applying the retained plus-congruence step-down to $\chi_A(2\ell) \equiv \chi_A(2p) \pmod{\chi_A(m)}$ gives $\ell \equiv \pm p \pmod{m}$, hence $R \equiv \pm p \pmod{c}$. The half-interval bounds force $p = R$.

Returning to $2n = R + \epsilon + bE$, the bound $E > R + 2$ rules out $b > 0$. If $\epsilon = -1$, then $p = 2n + 1$, and duplication with $\chi_A(2) > P_0$ gives $c > Ak > k(Y+1)$, contradicting $kY < c < k(Y+1)$. Therefore $N_k = N_t = 1$ and only now $C > 0$. The retained fixed-minus parity theorem yields $R \equiv 3 \pmod{4}$. All remaining lower-ratio, half-binomial, mask and native-compiler steps then have their original hypotheses. Exact sources and full proof-order details are in `PROOF.md` and the pinned `INHERITED_PELL_AUDIT.md`.

1.5 Completeness and the logical endpoint

From a finite ZFC refutation, the inherited normalized compiler supplies a positive tuple for this entire selected coefficient tuple. Keep all its witnesses except replace i_{old} by $i_{\mathrm{new}} = \Delta i_{\mathrm{old}}$. Then

$$(i_{\mathrm{new}} c^2)^2 = \Delta(f^2 - 1),$$

while K_{aux}, V and all outer factors are unchanged. Every required factor is $+1$, so both displayed equations vanish. Conversely the preceding soundness proof recovers a calibrated finite blank-K215 computation. Hence, in ordinary mathematics formalizable in ZFC,

$$\mathrm{ZFC} \vdash \left(\neg\exists \mathbf{x} \in \mathbb{Z}_{>0}^{15}\ P_{72}(\mathbf{x}) = 0\right) \longleftrightarrow \mathrm{Con}(\mathrm{ZFC}).$$

This is not a proof of consistency and not a new end-to-end proof-assistant certificate. It uses the inherited compiler and literal K215 calibration.

1.6 Exact degree, operation counts, and historical comparison

The kernel has 61 live producers: 34 multiplications and 27 additions/subtractions. Equation (1) adds three products, three residual subtractions, three squares and two additions: $72 = 40M + 32A$. Its residual degrees are $40, 37, 22$. The companion has $74 = 40M + 34A$, with degrees $22, 28, 22, 27$.

The main norm's cancellation is an ordinary polynomial identity:

$$(ac + Z)^2 - (a^2 + H)c^2 = 2acZ + Z^2 - Hc^2, \qquad Z = X + \gamma H.$$

The independent auditor expands that cone exactly before propagating leading homogeneous forms. The two polynomial leaders are respectively

$$64\gamma^2 w^8 s^{14} (\eta + \zeta)^6 (\beta J)^{50}, \qquad w^4 s^8 T^4 f^8 (\eta + \zeta)^4 (\beta J)^{28}.$$

Their nonzero coefficients $64\beta^{50}$ and β^{28} prove exact degrees 80 and 56 for the actual positive fixed β.

The historical pairs (degree, operations) $(99, 69), (93, 70), (83, 71), (56, 74), (44, 78)$ remain relevant. The present $(80, 72)$ point does not dominate $(83, 71)$ or $(56, 74)$; it lies between them. In particular, $(56, 74)$ is *not* a new record here. The prior relaxed kernel and its historical Q_{56} are explicitly retained for comparison. The reproducible standard-library audit verifies counts, liveness, literal sources, complete payloads and exact degree, but does not itself certify the unbounded compiler semantics.